Lecture 9 Tutorial — Answers

Q1. Confusion Matrix Entries

1. Predicted positive, actually positive: **TP**
2. Predicted positive, actually negative: **FP**
3. Predicted negative, actually positive: **FN**
4. Predicted negative, actually negative: **TN**

Q2. Metrics from Confusion Matrix

1. Accuracy:
$$\text{Acc} = \frac{TP + TN}{TP + TN + FP + FN}$$
2. Recall (Sensitivity / TPR):
$$\text{Recall} = \frac{TP}{TP + FN}$$
3. Precision:
$$\text{Precision} = \frac{TP}{TP + FP}$$
4. Specificity (TNR):
$$\text{Specificity} = \frac{TN}{TN + FP}$$
5. Fall-out (FPR):
$$\text{Fallout} = \frac{FP}{TN + FP}$$
6. Miss rate (FNR):
$$\text{Miss} = \frac{FN}{TP + FN}$$

Q3. Complementary Rates

1.
$$\text{Recall} + \text{Miss} = \frac{TP}{TP + FN} + \frac{FN}{TP + FN} = \frac{TP + FN}{TP + FN} = 1$$
2.
$$\text{Specificity} + \text{Fallout} = \frac{TN}{TN + FP} + \frac{FP}{TN + FP} = \frac{TN + FP}{TN + FP} = 1$$

Q4. Compute Metrics for Given Cases

Use
$$F_1 = \frac{2TP}{2TP + FP + FN}.$$

Case A: $(63, 28, 37, 72)$

$$\text{Accuracy} = \frac{63 + 72}{200} = 0.675$$

$$\text{Specificity} = \frac{72}{72 + 28} = 0.720$$

$$\text{Recall} = \frac{63}{63 + 37} = 0.630$$

$$\text{Precision} = \frac{63}{63 + 28} = \frac{63}{91} = 0.692$$

$$F_1 = \frac{2 \cdot 63}{2 \cdot 63 + 28 + 37} = \frac{126}{191} = 0.660$$

Case B: $(77, 77, 23, 23)$

$$\text{Accuracy} = \frac{77 + 23}{200} = 0.500$$

$$\text{Specificity} = \frac{23}{23 + 77} = 0.230$$

$$\text{Recall} = \frac{77}{77 + 23} = 0.770$$

$$\text{Precision} = \frac{77}{77 + 77} = 0.500$$

$$F_1 = \frac{2 \cdot 77}{2 \cdot 77 + 77 + 23} = \frac{154}{254} = 0.606$$

Case C: $(24, 88, 76, 12)$

$$\text{Accuracy} = \frac{24 + 12}{200} = 0.180$$

$$\text{Specificity} = \frac{12}{12 + 88} = 0.120$$

$$\text{Recall} = \frac{24}{24 + 76} = 0.240$$

$$\text{Precision} = \frac{24}{24 + 88} = \frac{24}{112} = 0.214$$

$$F_1 = \frac{2 \cdot 24}{2 \cdot 24 + 88 + 76} = \frac{48}{212} = 0.226$$

Case D: $(76, 12, 24, 88)$

$$\text{Accuracy} = \frac{76 + 88}{200} = 0.820$$

$$\text{Specificity} = \frac{88}{88 + 12} = 0.880$$

$$\text{Recall} = \frac{76}{76 + 24} = 0.760$$

$$\text{Precision} = \frac{76}{76 + 12} = \frac{76}{88} = 0.864$$

$$F_1 = \frac{2 \cdot 76}{2 \cdot 76 + 12 + 24} = \frac{152}{188} = 0.809$$

Q5. F1-score Equivalence

$$
\begin{aligned}
F_1 &= \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \\
&= \frac{2\left(\frac{TP}{TP+FP}\right)\left(\frac{TP}{TP+FN}\right)}{\left(\frac{TP}{TP+FP}\right)+\left(\frac{TP}{TP+FN}\right)} \\
&= \frac{2TP^2}{(TP+FP)(TP+FN)} \cdot \frac{(TP+FP)(TP+FN)}{TP(TP+FN)+TP(TP+FP)} \\
&= \frac{2TP^2}{TP(2TP+FP+FN)} \\
&= \frac{2TP}{2TP+FP+FN}
\end{aligned}
$$

Q6. Thresholding a Score-Based Classifier

Ground-truth positives: items $1, 4, 6, 9, 10$.

Ground-truth negatives: items $2, 3, 5, 7, 8$.

(1) For $\tau = 0.50$

Predicted positive: items $3, 4, 6, 8, 9$.

So

$$TP = 3, \quad FP = 2, \quad FN = 2, \quad TN = 3.$$

Metrics:

$$
\begin{aligned}
\text{Accuracy} &= \frac{3+3}{10} = 0.600 \\
\text{Recall} &= \frac{3}{3+2} = 0.600 \\
\text{Precision} &= \frac{3}{3+2} = 0.600 \\
F_1 &= \frac{2 \cdot 3}{2 \cdot 3 + 2 + 2} = \frac{6}{10} = 0.600
\end{aligned}
$$

(2) For $\tau = 0.70$

Predicted positive: items $3, 6, 9$.

So

$$TP = 2, \quad FP = 1, \quad FN = 3, \quad TN = 4.$$

Metrics:

$$
\begin{aligned}
\text{Accuracy} &= \frac{2+4}{10} = 0.600 \\
\text{Recall} &= \frac{2}{2+3} = 0.400 \\
\text{Precision} &= \frac{2}{2+1} = 0.667 \\
F_1 &= \frac{2 \cdot 2}{2 \cdot 2 + 1 + 3} = \frac{4}{8} = 0.500
\end{aligned}
$$

(3) Higher F_1

$$\boxed{\tau = 0.50}$$

Q7. ROC Operating Points

Using

$$\text{FPR} = \frac{FP}{FP + TN}, \qquad \text{TPR} = \frac{TP}{TP + FN}.$$

τ	FPR	TPR
0.90	$1/5 = 0.200$	$0/5 = 0.000$
0.70	$1/5 = 0.200$	$2/5 = 0.400$
0.50	$2/5 = 0.400$	$3/5 = 0.600$
0.30	$4/5 = 0.800$	$4/5 = 0.800$

Q8. AUC

A. AUC is the area under the ROC curve.

Q9. Cross-Validation Methods

A–2, B–1, C–3, D–4

Q10. k-fold Mechanics

A dataset has $n = 103$ samples and $k = 10$ folds.

1. Number of models trained:

$$\boxed{10}$$

2. Since

$$103 = 10 \times 10 + 3,$$

there are **3 folds of size 11** and **7 folds of size 10**.

3. Final reported performance:

$$\frac{1}{10} \sum_{i=1}^{10} m_i$$

Q11. Three-way Split and Leakage

B. Because using the validation set to select the final model biases the error estimate downward; the test set is reserved for final assessment only.

Q12. Model Selection Criteria: AIC and BIC

Given

$$AIC = 2k - 2\ln(L), \qquad BIC = k\ln(n) - 2\ln(L), \qquad n = 1000.$$

For Model 1:

$$k = 5, \qquad \ln(L) = -1200$$

$$AIC_1 = 2(5) - 2(-1200) = 10 + 2400 = 2410$$

$$BIC_1 = 5\ln(1000) - 2(-1200) = 5\ln(1000) + 2400$$

Using

$$\ln(1000) \approx 6.908,$$

$$BIC_1 \approx 5(6.908) + 2400 = 34.540 + 2400 = 2434.540$$

For Model 2:

$$k = 12, \qquad \ln(L) = -1180$$

$$AIC_2 = 2(12) - 2(-1180) = 24 + 2360 = 2384$$

$$BIC_2 = 12\ln(1000) - 2(-1180) = 12\ln(1000) + 2360$$
$$BIC_2 \approx 12(6.908) + 2360 = 82.896 + 2360 = 2442.896$$

1.
$$AIC_1 = 2410, \qquad AIC_2 = 2384$$

2.
$$BIC_1 \approx 2434.540, \qquad BIC_2 \approx 2442.896$$

3. Preferred model:

- Under AIC: **Model 2**
- Under BIC: **Model 1**